

ITI123 Project Milestone Report

AI-Based Badminton Stroke Classification Using Pose Estimation

Paul George Karippaparambil

8031408E

ITI123

January 16, 2026

1 Introduction

I play badminton recreationally. Getting technique feedback without a coach is hard. This project tests whether AI can classify two overhead shots using only body pose data:

- **Clear:** Hit shuttlecock high and deep
- **Smash:** Attack shot going steeply downward

Research question: Can skeletal keypoint data from video distinguish these strokes?
No racket tracking. No shuttlecock tracking. Just body pose.

1.1 Dataset

I used ShuttleSet—video clips from professional matches.

Numbers:

- Started with 4,983 clips
- Removed 301 backhand clips
- 4,682 forehand clips (2,371 Clear, 2,311 Smash)
- MediaPipe extracted poses from 99.4% (4,655 successful)
- From 40 different matches

Class balance: 50.6% Clear, 49.4% Smash.

1.2 Pipeline

Six steps:

1. Pose extraction using MediaPipe
2. Feature engineering (60 features per frame)

3. Group-stratified data splitting
4. Model training (multiple architectures)
5. Statistical validation
6. Results analysis

2 Technical Implementation

2.1 Pose Extraction

MediaPipe Pose extracts 33 body keypoints per frame with x, y, z coordinates. I configured it for highest accuracy (model complexity 2).

Key landmarks: shoulders, elbows, wrists, hips, knees, ankles, nose.

2.2 Feature Engineering

Raw coordinates aren't enough. I created 60 features per frame:

Table 1: Feature Categories

Category	Description
Arm Extension	Distance from shoulder to wrist
Depth	Wrist/elbow depth relative to shoulder/hip
Height	Wrist/elbow height relative to shoulder/head
Joint Angles	Elbow angle, shoulder angle
Body Posture	Torso lean, shoulder rotation
Velocities	First derivative of all spatial features
Wrist/Forearm Features (9 features):	
r forearm vertical angle	Angle between forearm and vertical
r wrist elbow vertical	Wrist height above elbow
r forearm pitch	Forearm elevation angle
+ 6 more	Pronation, horizontal reach, arm plane angles

Hypothesis: Wrist angle would discriminate strokes. Clear uses extended wrist (120–150°). Smash uses flexed wrist (30–60°).

2.3 Data Splitting

Random splitting causes leakage. I used group-stratified splitting—all clips from same match stay together.

Results:

Table 2: Data Split Statistics

Split	Clips	Matches	% Clear	% Smash
Train	3,554	27	50.45	49.55
Validation	426	5	50.70	49.30
Test	675	8	54.22	45.78
Total	4,655	40	50.62	49.38

2.4 Padding and Normalization

Videos have different lengths. I padded to 50 frames. To avoid corrupting normalization, I computed statistics from real frames only, then padded afterward.

Models use Masking layers to ignore padding during training.

2.5 Model Architectures

I tested four baseline models:

- **ResNet1D**: Pattern detector across motion sequence
- **LSTM**: Memory-based, remembers motion history
- **BiLSTM**: Processes sequence forward and backward
- **GRU**: Simpler, faster version of LSTM

Training config:

- Optimizer: Adam (lr 0.001)
- Loss: Binary crossentropy
- Batch size: 32
- Max epochs: 50, early stopping (patience 10)

All models take input: 50 timesteps $\times$ 60 features.

3 Results

3.1 Baseline Models

Table 3: Baseline Model Results (Test Set)

Model	Accuracy	Precision	Recall	F1
ResNet1D	0.491	0.487	0.507	0.497
LSTM	0.457	0.459	0.446	0.452
BiLSTM	0.433	0.428	0.476	0.451
GRU	0.450	0.463	0.504	0.483
Random	0.500	0.500	0.500	0.500

Every model performed at or below random guessing. 50% accuracy means models learned nothing.

3.2 Enhanced Models

I tested whether more sophisticated models could find patterns:

Enhancements:

- Attention mechanisms
- Transformer architecture
- Data augmentation (time shifts, scaling, noise, mixup)
- Squeeze-and-Excitation blocks
- Label smoothing

Table 4: Enhanced Model Results (Test Set)

Model	Accuracy	Precision	Recall	F1
ResNet Enhanced	0.51	0.51	0.50	0.50
LSTM + Attention	0.50	0.50	0.51	0.50
BiLSTM + Attention	0.49	0.49	0.50	0.49
Transformer	0.50	0.50	0.49	0.50
Random	0.50	0.50	0.50	0.50

No improvement. All models converge to 50% regardless of complexity.

Conclusion: Problem is data, not models.

4 Analysis

4.1 Why Classification Failed

Dataset captures shuttlecock contact moment only. At this instant, clear and smash have identical body poses:

Table 5: Body Position at Contact

Feature	Clear	Smash
Racket position	Overhead	Overhead
Contact height	2m above court	2m above court
Arm extension	Fully extended	Fully extended
Body posture	Upright	Upright
Forearm angle	$85.3^\circ \pm 45.2^\circ$	$84.2^\circ \pm 44.8^\circ$
Difference	1.1° (negligible)	

My wrist angle hypothesis was wrong. Expected 60–90° difference. Got 1.1°.

4.2 Missing Information

Key discriminators aren't in pose data:

1. **Racket face angle:** MediaPipe doesn't track equipment. Clear angles racket upward. Smash angles downward. Most important feature—missing.
2. **Shuttlecock trajectory:** Clears go high. Smashes go steep. Clips end at contact. No trajectory.
3. **Follow-through:** Clears continue upward. Smashes snap downward. Clips end at contact. No follow-through.
4. **Racket velocity:** Smashes have higher speeds. Body velocities don't capture this.

4.3 What Worked

Despite failure, methodology was sound:

- Group-stratified splitting prevented data leakage
- Proper padding and normalization maintained data integrity
- Multiple architectures validated conclusion
- Early failure prevented wasted effort on model tuning

5 Recommendations

5.1 In the next phase

Four approaches to make this work:

1. **Different strokes:** Add different stroke like lift or backhand to improve classification
2. **Improve Feature engineering:** Add more feature engineering ideas from coaching perspective that can detect the shot
3. **Extend time window:** Include 2–3 seconds after contact. Capture follow-through. Medium impact.

6 AI Governance

I evaluated against Singapore's AI Governance Framework (ISAGo).

6.1 Key Principles

Transparency: Users see exact measurements and benchmark sources.

Privacy: On-device processing. Immediate video deletion. No tracking.

User Control: System suggests only. Never commands.

Before public release:

- Add disclaimers
- Test fairness across demographics
- Handle edge cases
- Security audit

7 Conclusion

This project tested whether body pose alone distinguishes badminton clear and smash strokes.

Answer: No.

Key finding: Clear and smash have identical body poses at contact. Difference: 1.1° in forearm angle. Models achieved 50% accuracy—equivalent to random guessing.

What I built:

- Complete ML pipeline: Raw match videos to clip extraction and pose extraction
- 60 engineered features including 9 domain-specific wrist features
- Proper data handling: group-stratified split, padding masking
- Multiple architectures: baselines to Transformers

Next steps: Success requires (1) Different model evaluation and (2) different stroke pairs with distinct body mechanics.

Acknowledgments

This project used ShuttleSet dataset [1] and MediaPipe from Google [2]. Code development, debugging, Report drafting and Latex Formatting was assisted by Claude Code [3] and ChatGPT [4].

References

- [1] Wei-Yao Wang et al. *ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis*. CoRR, abs/2306.04948, 2023.
- [2] Google Research. *MediaPipe Pose*. <https://google.github.io/mediapipe/solutions/pose.html>, 2023.
- [3] Anthropic. *Claude Code: AI-Powered Command Line Coding Tool*. <https://www.anthropic.com/claude/code>, 2025.
- [4] OpenAI. *ChatGPT: Optimizing Language Models for Dialogue*. <https://openai.com/chatgpt>, 2024.